0006 — QuantSystems "Breakouts & Reversals" — Reproduction & Edge Study — 2026-06-24

Series: MC Setup Research Notes · Note 0006 **Confidence:** High (on the verdict). A 5-year, tick-level, full-cost reproduction of Ali Moin-Afshari's actual PaintBar code (TradeStation Ver 2 & Ver 5, cross-checked against his Pine v7.6) **and** his whitepaper's page-8/9 research definitions. The signal detection is faithful (our per-day frequencies match the paper); the edge verdict is firmly negative and is corroborated by his own source video and research worksheet.

TL;DR. We set out to reproduce and test Ali Moin-Afshari's breakout/reversal setups (Breakouts Research Whitepaper, Rev 3). We obtained his real **PaintBar source** and his **research worksheet + a mentoring video**, so this is a faithful decode, not a guess. **The signal detection reproduces** — the bars his indicator paints (which Al Brooks personally validated) — and our mechanical detector's per-day counts match his paper (BO+FT ~5/day, Rev+FT ~6.6/day). **But traded mechanically over 5 unselected years (2021-06->2026-06), the setups have no edge:** ~48–49% win, ~0 to slightly negative expectancy even **frictionless**, and negative after costs. His headline numbers (SQN 6.3–14.9, win 76–99%) come from **100 hand-picked 2020 trades with discretionary filtering**. An early in-app **+$359k/contract was an intrabar look-ahead bug** (caught and fixed; the edge vanished with it). His own video is explicit: the system is **discretionary second-leg trading with pullback scale-in and "ignore the questionable"** — "the computer marks the bars… but you still have to judge — that's your experience." **Verdict: the signals are reproducible; the edge is discretionary and not mechanizable.**

1. What this is, and the sources (so it stands alone)

- **Instrument/timeframe:** ES 5-minute, RTH. ES = $50/pt, 1 tick = $12.50.
- **Three primary sources, in order of authority for the rules:**

1. **PaintBar AMA_Breakouts_PB** EasyLanguage source — **Ver 2 (Aug 2022)** and **Ver 5 (Feb 2023)** — plus the **Pine v7.6** port. These agree on the core logic; the Pine is the most recent and adds a Z-score "big breakout." This is the ground truth for what a signal is. 2. **The whitepaper (Rev 3, Nov 2023)** — pages 8–9 give the research-sampling definitions (BO+FT, Big Breakout, Reversal+FT) used to produce the SQN stats. These differ slightly from the paint code (see §3.3). 3. **A mentoring video + his research worksheet** (Dev_BO Detector_PB, AMA_Breakout Detector_PB_X v11X; ES.D, 24-Aug->20-Nov-2020). These reveal the trading method and the provenance of the numbers — decisive for the verdict (§6).

- **What "reproduce" means here:** a pure, headless, sweepable Python detector

(qs_setups.py) that emits the project-standard signal schema, fed through the audited tick-level engine (simulation_engine.simulate_trades) with the real cost model — the same pipeline as every other note in this series.

2. The questions

1. Can the signals be defined precisely enough to reproduce mechanically? (Yes.) 2. Do they carry a tradeable edge on unselected, out-of-2020 data with realistic execution? 3. If the paper shows SQN 6–15, where does that actually come from?

3. The signals — precise, codeable definitions

3.1 Building blocks (verbatim from the code)

- **IBS** = (Close − Low) / (High − Low) x 100. Degenerate bar (H=L) -> 50.
- **BarDir** (±1): the paper's exact cascade — C vs O combined with IBS-vs-50; dojis and

body/close conflicts resolve by close; perfect dojis & ambiguous bars inherit the prior bar.

- **ABR / AvgRange:** average bar range. Two conventions exist: the whitepaper uses **prior**

10 bars **("within the last 10 bars"); the Ver 5 range filter uses** Average(range, 8) including the current bar**. Both are exposed as parameters.**

- **Outside bar (OB):** H>H[1] & L<L[1] (plus two equal-edge variants unless _StrictOB).
- **Inside bar (IB):** contained by the prior bar (+ equal-edge variants).
- **Micro gap (MIG):** gap between two non-adjacent bars (3-bar). Tagged, not gated.

3.2 PaintBar primitives (what the indicator actually marks)

Priority **CX > OB > BO**. (Ver 5 geometry; >=/<= and OB-gated.)

- **BO (breakout):** bull = H>H[1] AND L>=L[1] and **not** an OB; bear = H<=H[1] AND L<L[1]

and not OB. **No IBS / ABR / FT in the bare paint signal** — those are optional filters.

- **OB (outside-bar) directional bias:** compare breakout above prior high (H−H[1]) vs below

prior low (L[1]−L); bigger-above => bullish bias; tie & perfect-doji => neutral (magenta).

- **CX (climax) — _CXfactor resolved = 1.8** (placeholder P3 from the first draft, now known):

on a clean directional BO bar, HHdist > LLdist x 1.8 (a thrust), filtered out if the prior bar is inside, the bar is an OB, BarDir≠BarDir[1], or it doesn't close beyond the prior extreme.

- **Range filter (Ver 5 default ON, lookback 8):** drop a signal whose bar range < AvgRange.
- **Follow-through (_PaintFTBar, default OFF):** relabels a bar as FT when the prior bar was a signal and this bar closes beyond it; also filters mid-leg non-close-beyond breakouts.

3.3 Whitepaper page-8/9 RESEARCH setups (the SQN study's definitions)

These are a layer on top of the primitives, and are what the paper's stats describe. Built as detect_wp():

- **1. BO+FT (2-bar):** breakout bar (1 tick beyond the prior bar) + a **same-direction** FT bar (need not close beyond); **>=1 of the two bars > ABR(10)**; breakout-bar **IBS >=69 / <=31**.
- **2. Big Breakout:** a bar **> 2x ABR(10)**, breakout, strong IBS (1-bar or 2-bar).
- **3. Reversal+FT (2-bar):** both bars >= ABR(10) (not breakout-big), **bar 2 closes beyond** bar 1**, at a reversal turn, signal-bar** IBS >=69 / <=31. **(Item 3's "not big enough to be a breakout bar" is self-contradictory in the paper (A4); we relaxed it to match his count.)**

Note: OB and CX have **no mechanical rules in the whitepaper** ("I have not provided detailed rules for the last two signals") — those live only in the paint code. And the shipped detector is **version 24 with ~60 price-action rules** (some firing once a quarter); the Ver5/Pine we coded is a faithful **subset.**

3.4 iStop and target (corrected from the worksheet)

- **iStop = 1x the signal-bar range** (R = entry − stop). His worksheet's actual-risk column (a.Risk Dist) averaged **~4.74 pt** in 2020 low-vol ES ~ one bar range — not the 2x-combined-range rule the first draft assumed. (Variants for small / much-bigger bars exist.)
- **Target "1R" = a second leg ~ the size of the signal bar** (his "symmetry") -> roughly symmetric 1:1 with the 1x stop.

4. Entry methodology (whitepaper §"Trading Techniques" + Fig 12–13, p. 22)

Entry is **off the signal bar**, three techniques: 1. **Buy-The-Close (BTC)** — his preferred (limit) style: enter at the **close of the signal bar** — either **B2** (setup complete / FT bar, confirmed) or **B1** (early, before FT). 2. **Market** — market order at the signal-bar close. 3. **Scaling-In** — add with **limit orders on the pullback** at prior support/resistance.

Stop = iStop. Target = 1R (symmetric second leg). The R-multiple study measured two: **(a) entry bar 2, hold 1R** -> BO SQN 6.3 / Rev SQN 4.4; **(b) entry bar 1, add on the pullback (scale-in), look for 1R** -> BO SQN 10.2 / Rev SQN 9.2 — **(b) is his real, preferred method.**

5. How we reproduced & tested it

- **Detection:** qs_setups.py — detect() (PaintBar primitives) and detect_wp() (whitepaper setups); every parameter on a QSConfig dataclass (fully sweepable). Presets: wp(), paper(), research(), paintbar_raw().
- **Simulation:** simulation_engine.simulate_trades — replays **real continuous ticks** per day, R = |entry−stop|, exits at target_r·R or stop, house costs ($5 RT + slip), ESA execution models. Entry = market at signal-bar close (~ BTC).
- **In-app:** QS Breakouts tab (detector selector, geometry, ESA, BE, Quick View, Y/Q/M/W breakdowns, trade list, CSV export) -> results push to ba_results so the **Prop Sim** tab runs on QS trades. NT indicator QSBreakoutSignals.cs paints the exported signals on a chart.

6. Results

6.1 Frequency — detection is faithful [OK]

detect_wp over 1,249 RTH sessions vs the paper:

Setup	ours (ok/day)	ours (raw/day)	paper
BO+FT	**5.2**	8.0	~5/day
Rev+FT	4.2	**6.1**	~6.6/day
BigBO	1.0	1.5	(subset)

Right order on every setup — the signal definitions reproduce his population.

6.2 The look-ahead bug (the headline cautionary tale)

An early in-app run showed **+$359k/contract, 78% win, SQN 10** — exciting, and **wrong.** The detector timestamped each signal at the bar's **open**; the engine then filled at the **first tick after the open** = inside the signal bar, ~5 min before the signal exists. 100% of fills were intra-signal-bar, always favorable (longs filled near the low, shorts near the high). **Fixed** by emitting the bar **close** time. The "edge" was entirely this leak.

6.3 Honest edge — WP setups, correct geometry, look-ahead-free, **frictionless**

Setup @ target	n	win%	exp R	SQN	net $ (1 ES)
BO+FT @ 1R	5,692	48.6%	**−0.026**	−2.0	−$14k
BO+FT @ 2R	5,692	34.6%	−0.013	−0.7	+$16k
Rev+FT @ 1R	5,144	48.5%	**−0.028**	−2.1	−$69k
Rev+FT @ 2R	5,144	35.2%	−0.020	−1.1	−$33k
Ali (2020, 100 hand-picked)	100	**76%**	+0.60	6.3	—

Coin-flips, slightly **negative even before costs** -> firmly negative after. No mechanical edge. (Trade-level audit confirmed clean execution: 0 entry-bar target wins, median ~17-bar holds, fills strictly after the signal close.)

7. Why it fails mechanically — and where Ali's edge actually lives

His own materials make this unambiguous, so this is not speculation:

- **It's discretionary second-leg trading, not a signal-bar system.** Video: "the only trade you have to look at is second legs… >90% chance… that's why I programmed this breakout detector — it's designed for second-leg trading." The detector marks the **first** leg; the **trade** is the **second leg after a pullback**, entered by **scaling in** during the pullback.
- **The edge is the "ignore" judgment.** Video (the Heat analogy): "disqualify the questionable… if it's questionable for any reason, just ignore it." He then rejects setups by eye (3rd legs, a red bar on the MA, wedge/magnet targets, range-compression breakouts). And: "the computer marks the bars… but you still have to judge — that's your experience."
- **The numbers are 100 hand-picked 2020 trades + discretionary filters.** Worksheet: 100 trades, done **by hand** ("you want to live through the trades"), Aug–Nov 2020 (a trending regime). The **filter-ablation table** shows the lift comes from removing **legs 3 & 4** (76%->83%) and **time-of-day** (->86%) — and "legs 3 & 4" is a **hand-labeled column** (his channel read), i.e. exactly the part that **can't be mechanized**.
- **Brooks's endorsement was about the bars, not an edge:** "it's picking up the bars I'm interested to trade." That's signal detection — which we reproduced — not profitability.

8. Recommendation / verdict

- **The signals are faithfully reproducible** (detection [OK]) — useful as a charting/marking layer and as a research substrate.
- **There is no standalone mechanical edge** in BO+FT or Rev+FT on 5 unselected years, at any of the geometries tested, frictionless or with costs.
- **Do not trade these as a mechanical signal-bar system.** The reported SQN 6–15 is a discretionary, hand-curated, single-regime result; treat it as a description of a skilled manual trader's process, not a backtestable system.

9. Caveats & open questions

- **Not yet tested:** the faithful version of his method — **bar-1 entry + pullback scale-in,** symmetric second-leg target **(his SQN-10.2 line). Prior: note 0001 found PB scale-in helps win% but not $ on the MC setup; expectation is similar here, but it's the honest final check.**
- **Legs-3/4 + deep-PB + "ignore" filters** are discretionary and unmechanizable; we can only approximate them, and approximations won't recover a hand-trader's judgment.
- The full detector is **~60 rules (v24)**; we reproduced the documented subset. A fuller port would mark more bars but doesn't change the edge conclusion (the edge isn't in detection).
- In-sample on 2021–2026; **2020 (his sample) is not in our tick set** (our data starts 2021-06-18), so a direct match against his hand-logged trades isn't possible yet. **Strongest outstanding validation:** fetch Aug–Dec 2020 ES, run detect_wp on his exact worksheet dates/times, and check signal-by-signal whether his logged setups are a subset of our detections. Ground-truth seed (partial, transcribed from the worksheet frames) saved at docs/living/ali_2020_worksheet_trades.csv; full numeric extraction needs the actual .xlsx.

10. Reproduce (artifacts)

- **Detector:** qs_setups.py — detect(), detect_wp(), QSConfig (+ presets).
- **Scripts:** scripts/qs_breakouts_detect.py (frequency), qs_config_compare.py, qs_ali_geometry.py, qs_breakouts_sim.py, qs_wp_sim.py, qs_breakouts_periods.py, qs_breakouts_be_sweep.py,

qs_stop_target_sweep.py, qs_dump_trades.py, qs_export_signals_nt.py.

- **App:** qs_tab.py (QS Breakouts tab) -> pushes to Prop Sim.
- **NT:** QSBreakoutSignals.cs (racing-stripe signal marker) + qs_signals_.csv.
- **Sources:** whitepaper Rev 3; PaintBar Ver 2 / Ver 5 / Pine v7.6; mentoring video + worksheet.
- **Companion:** note 0007 (build/test mechanism); the inflated pre-fix numbers there are voided

by §6.2 above.